

frisk.r

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```
setwd("~/Dropbox/Classes/536/CaseStudies/PoliceStops")

frisk <- read.table("frisk.dat",header=TRUE)
attach(frisk)

n <- length(unique(precinct))*length(unique(eth))
mystops <- numeric(n)
myarrests <- numeric(n)
myprecinct <- numeric(n)
myeth <- numeric(n)

k <- 1
for(i in 1:length(unique(precinct)))
{
  for(j in 1:length(unique(eth)))
  {
    myprecinct[k] = i
    myeth[k] = j
    mystops[k] = sum(frisk[(precinct==i) & (eth==j),1])
    myarrests[k] = sum(frisk[(precinct==i) & (eth==j),3])
    k = k+1
  }
}

mydata <- cbind(mystops,myarrests,myprecinct,myeth)
write.table(mydata,file="policestops.txt",row.names=FALSE,col.names=TRUE)
#####
data <- read.table("policestops.txt",header = TRUE)
attach(data)

## The following objects are masked _by_ .GlobalEnv:
##
##      myarrests, myeth, myprecinct, mystops

model1A <- glm(mystops ~ 1,family=poisson(link = "log"))
summary(model1A)

##
## Call:
## glm(formula = mystops ~ 1, family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -33.049  -22.552   -8.788    9.343   65.227
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept) 6.370053 0.002758 2309 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 123333 on 224 degrees of freedom
## Residual deviance: 123333 on 224 degrees of freedom
## AIC: 125041
##
## Number of Fisher Scoring iterations: 5
model2A <- glm(mystops ~ factor(myeth),family=poisson(link = "log"))
summary(model2A)

##
## Call:
## glm(formula = mystops ~ factor(myeth), family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -42.184  -16.723   -6.891    8.867   57.213
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    6.836231   0.003784 1806.41  <2e-16 ***
## factor(myeth)2 -0.447714   0.006061  -73.87  <2e-16 ***
## factor(myeth)3 -1.414281   0.008558 -165.26  <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 123333 on 224 degrees of freedom
## Residual deviance: 88771 on 222 degrees of freedom
## AIC: 90484
##
## Number of Fisher Scoring iterations: 5
model3A <- glm(mystops ~ factor(myeth)+factor(myprecinct),family=poisson(link = "log"))
summary(model3A)
```

```
##
## Call:
## glm(formula = mystops ~ factor(myeth) + factor(myprecinct), family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -36.603  -12.500   -2.412    8.935   38.919
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    5.320809   0.051031  104.267  < 2e-16 ***
## factor(myeth)2  -0.447714   0.006061  -73.872  < 2e-16 ***
## factor(myeth)3  -1.414281   0.008558 -165.263  < 2e-16 ***
```

```

## factor(myprecinct)2 -0.103919 0.074022 -1.404 0.160352
## factor(myprecinct)3 1.426389 0.056756 25.132 < 2e-16 ***
## factor(myprecinct)4 1.298811 0.057499 22.588 < 2e-16 ***
## factor(myprecinct)5 1.424516 0.056766 25.094 < 2e-16 ***
## factor(myprecinct)6 1.214566 0.058038 20.927 < 2e-16 ***
## factor(myprecinct)7 0.522189 0.064329 8.117 4.76e-16 ***
## factor(myprecinct)8 1.406861 0.056864 24.741 < 2e-16 ***
## factor(myprecinct)9 -0.300754 0.078142 -3.849 0.000119 ***
## factor(myprecinct)10 1.099478 0.058843 18.685 < 2e-16 ***
## factor(myprecinct)11 0.777775 0.061569 12.633 < 2e-16 ***
## factor(myprecinct)12 0.831214 0.061062 13.613 < 2e-16 ***
## factor(myprecinct)13 1.749764 0.055217 31.689 < 2e-16 ***
## factor(myprecinct)14 1.107233 0.058786 18.835 < 2e-16 ***
## factor(myprecinct)15 1.818667 0.054944 33.101 < 2e-16 ***
## factor(myprecinct)16 0.918885 0.060280 15.244 < 2e-16 ***
## factor(myprecinct)17 0.898999 0.060452 14.871 < 2e-16 ***
## factor(myprecinct)18 1.812750 0.054966 32.979 < 2e-16 ***
## factor(myprecinct)19 1.225302 0.057967 21.138 < 2e-16 ***
## factor(myprecinct)20 1.403037 0.056886 24.664 < 2e-16 ***
## factor(myprecinct)21 1.432608 0.056722 25.257 < 2e-16 ***
## factor(myprecinct)22 2.348278 0.053344 44.022 < 2e-16 ***
## factor(myprecinct)23 1.802095 0.055008 32.761 < 2e-16 ***
## factor(myprecinct)24 1.918449 0.054578 35.150 < 2e-16 ***
## factor(myprecinct)25 2.258782 0.053561 42.172 < 2e-16 ***
## factor(myprecinct)26 1.386944 0.056977 24.342 < 2e-16 ***
## factor(myprecinct)27 1.650156 0.055643 29.656 < 2e-16 ***
## factor(myprecinct)28 0.851371 0.060877 13.985 < 2e-16 ***
## factor(myprecinct)29 2.015076 0.054256 37.140 < 2e-16 ***
## factor(myprecinct)30 1.723230 0.055326 31.147 < 2e-16 ***
## factor(myprecinct)31 1.545073 0.056138 27.523 < 2e-16 ***
## factor(myprecinct)32 0.994694 0.059649 16.676 < 2e-16 ***
## factor(myprecinct)33 2.041726 0.054172 37.690 < 2e-16 ***
## factor(myprecinct)34 1.987696 0.054344 36.576 < 2e-16 ***
## factor(myprecinct)35 0.631548 0.063076 10.012 < 2e-16 ***
## factor(myprecinct)36 1.127625 0.058638 19.230 < 2e-16 ***
## factor(myprecinct)37 0.951507 0.060003 15.858 < 2e-16 ***
## factor(myprecinct)38 1.546733 0.056129 27.557 < 2e-16 ***
## factor(myprecinct)39 1.856298 0.054802 33.873 < 2e-16 ***
## factor(myprecinct)40 0.802526 0.061331 13.085 < 2e-16 ***
## factor(myprecinct)41 1.637104 0.055702 29.390 < 2e-16 ***
## factor(myprecinct)42 1.939956 0.054504 35.593 < 2e-16 ***
## factor(myprecinct)43 1.452252 0.056615 25.651 < 2e-16 ***
## factor(myprecinct)44 1.466737 0.056538 25.942 < 2e-16 ***
## factor(myprecinct)45 2.037334 0.054185 37.599 < 2e-16 ***
## factor(myprecinct)46 2.190855 0.053739 40.769 < 2e-16 ***
## factor(myprecinct)47 0.955511 0.059970 15.933 < 2e-16 ***
## factor(myprecinct)48 1.408132 0.056857 24.766 < 2e-16 ***
## factor(myprecinct)49 0.994694 0.059649 16.676 < 2e-16 ***
## factor(myprecinct)50 2.070964 0.054082 38.293 < 2e-16 ***
## factor(myprecinct)51 1.143478 0.058525 19.538 < 2e-16 ***
## factor(myprecinct)52 1.704039 0.055408 30.755 < 2e-16 ***
## factor(myprecinct)53 1.150079 0.058479 19.667 < 2e-16 ***
## factor(myprecinct)54 0.949499 0.060020 15.820 < 2e-16 ***
## factor(myprecinct)55 1.085540 0.058946 18.416 < 2e-16 ***

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## factor(myprecinct)56 0.395896 0.065922 6.006 1.91e-09 ***
## factor(myprecinct)57 0.799027 0.061365 13.021 < 2e-16 ***
## factor(myprecinct)58 1.936965 0.054514 35.531 < 2e-16 ***
## factor(myprecinct)59 1.091662 0.058900 18.534 < 2e-16 ***
## factor(myprecinct)60 2.033262 0.054198 37.515 < 2e-16 ***
## factor(myprecinct)61 1.440020 0.056681 25.405 < 2e-16 ***
## factor(myprecinct)62 1.446155 0.056648 25.529 < 2e-16 ***
## factor(myprecinct)63 1.176054 0.058298 20.173 < 2e-16 ***
## factor(myprecinct)64 1.255357 0.057772 21.730 < 2e-16 ***
## factor(myprecinct)65 1.783064 0.055083 32.371 < 2e-16 ***
## factor(myprecinct)66 2.033262 0.054198 37.515 < 2e-16 ***
## factor(myprecinct)67 1.842403 0.054854 33.588 < 2e-16 ***
## factor(myprecinct)68 1.057971 0.059153 17.885 < 2e-16 ***
## factor(myprecinct)69 1.191164 0.058195 20.468 < 2e-16 ***
## factor(myprecinct)70 1.861156 0.054784 33.973 < 2e-16 ***
## factor(myprecinct)71 2.026780 0.054218 37.382 < 2e-16 ***
## factor(myprecinct)72 2.224272 0.053650 41.459 < 2e-16 ***
## factor(myprecinct)73 2.249239 0.053585 41.975 < 2e-16 ***
## factor(myprecinct)74 1.237433 0.057888 21.376 < 2e-16 ***
## factor(myprecinct)75 -0.178692 0.075517 -2.366 0.017969 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 123333 on 224 degrees of freedom
## Residual deviance: 54487 on 148 degrees of freedom
## AIC: 56348
##
## Number of Fisher Scoring iterations: 6
#####
model1B <- glm(mystops ~ log(myarrests),family=poisson(link = "log"))
summary(model1B)

##
## Call:
## glm(formula = mystops ~ log(myarrests), family = poisson(link = "log"))
##
## Deviance Residuals:
## Min 1Q Median 3Q Max
## -36.974 -10.015 -3.189 7.572 35.359
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept) 1.01639 0.02114 48.08 <2e-16 ***
## log(myarrests) 0.78165 0.00289 270.46 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 123333 on 224 degrees of freedom
## Residual deviance: 40662 on 223 degrees of freedom
## AIC: 42373

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```
##
## Number of Fisher Scoring iterations: 5
model2B <- glm(mystops ~ log(myarrests)+factor(myeth),family=poisson(link = "log"))
summary(model2B)

##
## Call:
## glm(formula = mystops ~ log(myarrests) + factor(myeth), family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -36.753   -9.350   -3.598    6.657   34.898
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.833455   0.024818   73.88  <2e-16 ***
## log(myarrests)  0.685161   0.003231  212.04  <2e-16 ***
## factor(myeth)2 -0.080633   0.006275  -12.85  <2e-16 ***
## factor(myeth)3 -0.577118   0.009540  -60.49  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 123333  on 224  degrees of freedom
## Residual deviance:  36609  on 221  degrees of freedom
## AIC: 38323
##
## Number of Fisher Scoring iterations: 5
model3B <- glm(mystops ~ log(myarrests)+factor(myeth)+factor(myprecinct),family=poisson(link = "log"))
summary(model3B)

##
## Call:
## glm(formula = mystops ~ log(myarrests) + factor(myeth) + factor(myprecinct),
##      family = poisson(link = "log"))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
##  -9.8713   -2.7384   -0.1496    2.3674   10.8421
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.064495   0.064115  -32.200  < 2e-16 ***
## log(myarrests)   1.102724   0.005818  189.540  < 2e-16 ***
## factor(myeth)2    0.042217   0.007142   5.911 3.40e-09 ***
## factor(myeth)3   -0.350663   0.010301  -34.042  < 2e-16 ***
## factor(myprecinct)2 -0.146206   0.074033  -1.975  0.04828 *
## factor(myprecinct)3  0.469445   0.056990   8.237  < 2e-16 ***
## factor(myprecinct)4  1.198924   0.057564  20.828  < 2e-16 ***
## factor(myprecinct)5  0.169204   0.057160   2.960  0.00307 **
## factor(myprecinct)6  1.144506   0.058049  19.716  < 2e-16 ***
## factor(myprecinct)7  0.192467   0.064352   2.991  0.00278 **
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## factor(myprecinct)8 -0.571810 0.057785 -9.895 < 2e-16 ***
## factor(myprecinct)9 0.571418 0.078308 7.297 2.94e-13 ***
## factor(myprecinct)10 0.354032 0.058958 6.005 1.92e-09 ***
## factor(myprecinct)11 0.644488 0.061593 10.464 < 2e-16 ***
## factor(myprecinct)12 1.187689 0.061109 19.436 < 2e-16 ***
## factor(myprecinct)13 0.981249 0.055407 17.710 < 2e-16 ***
## factor(myprecinct)14 0.566939 0.058866 9.631 < 2e-16 ***
## factor(myprecinct)15 0.976603 0.055136 17.713 < 2e-16 ***
## factor(myprecinct)16 0.775324 0.060306 12.856 < 2e-16 ***
## factor(myprecinct)17 -0.181497 0.060955 -2.978 0.00291 **
## factor(myprecinct)18 -0.043245 0.055868 -0.774 0.43890
## factor(myprecinct)19 0.183844 0.058520 3.142 0.00168 **
## factor(myprecinct)20 -0.188742 0.057550 -3.280 0.00104 **
## factor(myprecinct)21 0.290518 0.057236 5.076 3.86e-07 ***
## factor(myprecinct)22 1.087071 0.053873 20.178 < 2e-16 ***
## factor(myprecinct)23 0.472302 0.055550 8.502 < 2e-16 ***
## factor(myprecinct)24 1.254029 0.054728 22.914 < 2e-16 ***
## factor(myprecinct)25 0.773667 0.054183 14.279 < 2e-16 ***
## factor(myprecinct)26 -0.310127 0.057747 -5.370 7.85e-08 ***
## factor(myprecinct)27 1.925582 0.055696 34.573 < 2e-16 ***
## factor(myprecinct)28 -0.936341 0.061691 -15.178 < 2e-16 ***
## factor(myprecinct)29 1.017377 0.054617 18.628 < 2e-16 ***
## factor(myprecinct)30 0.397550 0.055850 7.118 1.09e-12 ***
## factor(myprecinct)31 1.664417 0.056181 29.626 < 2e-16 ***
## factor(myprecinct)32 1.430371 0.059739 23.944 < 2e-16 ***
## factor(myprecinct)33 0.983811 0.054514 18.047 < 2e-16 ***
## factor(myprecinct)34 1.477213 0.054407 27.151 < 2e-16 ***
## factor(myprecinct)35 0.861767 0.063234 13.628 < 2e-16 ***
## factor(myprecinct)36 1.613905 0.058835 27.431 < 2e-16 ***
## factor(myprecinct)37 1.443919 0.060039 24.050 < 2e-16 ***
## factor(myprecinct)38 1.776356 0.056213 31.600 < 2e-16 ***
## factor(myprecinct)39 0.312139 0.055896 5.584 2.35e-08 ***
## factor(myprecinct)40 1.559758 0.061515 25.356 < 2e-16 ***
## factor(myprecinct)41 1.916918 0.055718 34.404 < 2e-16 ***
## factor(myprecinct)42 0.985760 0.054803 17.987 < 2e-16 ***
## factor(myprecinct)43 0.409898 0.057153 7.172 7.39e-13 ***
## factor(myprecinct)44 0.783062 0.056982 13.742 < 2e-16 ***
## factor(myprecinct)45 0.679296 0.054979 12.356 < 2e-16 ***
## factor(myprecinct)46 0.497988 0.054490 9.139 < 2e-16 ***
## factor(myprecinct)47 1.151039 0.059998 19.185 < 2e-16 ***
## factor(myprecinct)48 0.259638 0.057561 4.511 6.46e-06 ***
## factor(myprecinct)49 1.114782 0.059694 18.675 < 2e-16 ***
## factor(myprecinct)50 0.724083 0.054832 13.206 < 2e-16 ***
## factor(myprecinct)51 0.235702 0.058914 4.001 6.31e-05 ***
## factor(myprecinct)52 0.363869 0.055942 6.504 7.80e-11 ***
## factor(myprecinct)53 0.534050 0.058608 9.112 < 2e-16 ***
## factor(myprecinct)54 0.416570 0.060202 6.920 4.53e-12 ***
## factor(myprecinct)55 0.422505 0.059197 7.137 9.52e-13 ***
## factor(myprecinct)56 0.928660 0.066030 14.064 < 2e-16 ***
## factor(myprecinct)57 1.606798 0.061509 26.123 < 2e-16 ***
## factor(myprecinct)58 1.524606 0.054622 27.912 < 2e-16 ***
## factor(myprecinct)59 1.088903 0.058925 18.479 < 2e-16 ***
## factor(myprecinct)60 0.688737 0.054852 12.556 < 2e-16 ***
## factor(myprecinct)61 1.245606 0.056801 21.929 < 2e-16 ***

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## factor(myprecinct)62 1.153892 0.056751 20.333 < 2e-16 ***
## factor(myprecinct)63 1.328419 0.058313 22.781 < 2e-16 ***
## factor(myprecinct)64 1.761533 0.057839 30.456 < 2e-16 ***
## factor(myprecinct)65 1.900627 0.055170 34.450 < 2e-16 ***
## factor(myprecinct)66 2.111117 0.054241 38.921 < 2e-16 ***
## factor(myprecinct)67 1.114422 0.055213 20.184 < 2e-16 ***
## factor(myprecinct)68 2.289384 0.059472 38.495 < 2e-16 ***
## factor(myprecinct)69 1.773278 0.058313 30.410 < 2e-16 ***
## factor(myprecinct)70 0.909801 0.055265 16.463 < 2e-16 ***
## factor(myprecinct)71 1.426888 0.054324 26.266 < 2e-16 ***
## factor(myprecinct)72 1.371740 0.054020 25.393 < 2e-16 ***
## factor(myprecinct)73 0.860623 0.054092 15.910 < 2e-16 ***
## factor(myprecinct)74 1.097276 0.058153 18.869 < 2e-16 ***
## factor(myprecinct)75 1.654239 0.075902 21.794 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 123332.5 on 224 degrees of freedom
## Residual deviance: 3103.8 on 147 degrees of freedom
## AIC: 4966.4
##
## Number of Fisher Scoring iterations: 4
#####
modell1 <- glm(mystops ~ 1,family=poisson(link = "log"),offset=log(myarrests))
summary(modell1)

##
## Call:
## glm(formula = mystops ~ 1, family = poisson(link = "log"), offset = log(myarrests))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -46.702  -8.437   0.020  10.132  40.795
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.587715   0.002758  -213.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 46120 on 224 degrees of freedom
## Residual deviance: 46120 on 224 degrees of freedom
## AIC: 47829
##
## Number of Fisher Scoring iterations: 5
modell2 <- glm(mystops ~ factor(myeth),family=poisson(link = "log"),offset=log(myarrests))
summary(modell2)

##

```

```
## Call:
## glm(formula = mystops ~ factor(myeth), family = poisson(link = "log"),
##      offset = log(myarrests))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -47.327  -7.740  -0.182   10.241   39.140
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.588086   0.003784 -155.40  <2e-16 ***
## factor(myeth)2  0.070208   0.006061  11.58  <2e-16 ***
## factor(myeth)3 -0.161581   0.008558  -18.88  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 46120  on 224  degrees of freedom
## Residual deviance: 45437  on 222  degrees of freedom
## AIC: 47150
##
## Number of Fisher Scoring iterations: 5
model3 <- glm(mystops ~ factor(myeth)+factor(myprecinct),family=poisson(link = "log"),offset=log(myarrests),summary(model3))

##
## Call:
## glm(formula = mystops ~ factor(myeth) + factor(myprecinct), family = poisson(link = "log"),
##      offset = log(myarrests))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -11.1396  -3.0895  -0.1934   2.0988  10.4185
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.378868   0.051019 -27.027  < 2e-16 ***
## factor(myeth)2   0.010188   0.006802   1.498 0.134190
## factor(myeth)3  -0.419001   0.009435 -44.409  < 2e-16 ***
## factor(myprecinct)2 -0.149050   0.074030  -2.013 0.044077 *
## factor(myprecinct)3  0.559955   0.056758   9.866  < 2e-16 ***
## factor(myprecinct)4  1.210636   0.057549  21.037  < 2e-16 ***
## factor(myprecinct)5  0.282865   0.056794   4.981 6.34e-07 ***
## factor(myprecinct)6  1.144204   0.058047  19.712  < 2e-16 ***
## factor(myprecinct)7  0.218173   0.064335   3.391 0.000696 ***
## factor(myprecinct)8 -0.390565   0.056868  -6.868 6.51e-12 ***
## factor(myprecinct)9  0.485112   0.078151   6.207 5.39e-10 ***
## factor(myprecinct)10 0.417068   0.058851   7.087 1.37e-12 ***
## factor(myprecinct)11 0.654417   0.061586  10.626  < 2e-16 ***
## factor(myprecinct)12 1.148698   0.061067  18.811  < 2e-16 ***
## factor(myprecinct)13 1.053509   0.055250  19.068  < 2e-16 ***
## factor(myprecinct)14 0.614115   0.058803  10.444  < 2e-16 ***
## factor(myprecinct)15 1.054904   0.054956  19.195  < 2e-16 ***
```

```

## factor(myprecinct)16 0.797715 0.060290 13.231 < 2e-16 ***
## factor(myprecinct)17 -0.049312 0.060478 -0.815 0.414864
## factor(myprecinct)18 0.131645 0.054981 2.394 0.016649 *
## factor(myprecinct)19 0.319164 0.057999 5.503 3.74e-08 ***
## factor(myprecinct)20 -0.040574 0.056928 -0.713 0.476016
## factor(myprecinct)21 0.410988 0.056805 7.235 4.65e-13 ***
## factor(myprecinct)22 1.211227 0.053399 22.683 < 2e-16 ***
## factor(myprecinct)23 0.599919 0.055064 10.895 < 2e-16 ***
## factor(myprecinct)24 1.319454 0.054600 24.166 < 2e-16 ***
## factor(myprecinct)25 0.913359 0.053596 17.041 < 2e-16 ***
## factor(myprecinct)26 -0.147614 0.057006 -2.589 0.009612 **
## factor(myprecinct)27 1.895485 0.055664 34.052 < 2e-16 ***
## factor(myprecinct)28 -0.764964 0.060915 -12.558 < 2e-16 ***
## factor(myprecinct)29 1.124710 0.054267 20.726 < 2e-16 ***
## factor(myprecinct)30 0.524652 0.055374 9.475 < 2e-16 ***
## factor(myprecinct)31 1.649498 0.056167 29.368 < 2e-16 ***
## factor(myprecinct)32 1.386276 0.059679 23.229 < 2e-16 ***
## factor(myprecinct)33 1.083192 0.054214 19.980 < 2e-16 ***
## factor(myprecinct)34 1.518972 0.054355 27.945 < 2e-16 ***
## factor(myprecinct)35 0.879407 0.063190 13.917 < 2e-16 ***
## factor(myprecinct)36 1.616306 0.058796 27.490 < 2e-16 ***
## factor(myprecinct)37 1.409430 0.060008 23.487 < 2e-16 ***
## factor(myprecinct)38 1.763813 0.056195 31.388 < 2e-16 ***
## factor(myprecinct)39 0.501081 0.054841 9.137 < 2e-16 ***
## factor(myprecinct)40 1.531984 0.061467 24.924 < 2e-16 ***
## factor(myprecinct)41 1.913898 0.055716 34.351 < 2e-16 ***
## factor(myprecinct)42 1.084217 0.054511 19.890 < 2e-16 ***
## factor(myprecinct)43 0.541787 0.056645 9.565 < 2e-16 ***
## factor(myprecinct)44 0.883038 0.056671 15.582 < 2e-16 ***
## factor(myprecinct)45 0.838439 0.054215 15.465 < 2e-16 ***
## factor(myprecinct)46 0.655855 0.053750 12.202 < 2e-16 ***
## factor(myprecinct)47 1.128144 0.059981 18.808 < 2e-16 ***
## factor(myprecinct)48 0.411880 0.056895 7.239 4.51e-13 ***
## factor(myprecinct)49 1.101219 0.059681 18.452 < 2e-16 ***
## factor(myprecinct)50 0.878797 0.054109 16.241 < 2e-16 ***
## factor(myprecinct)51 0.348980 0.058550 5.960 2.52e-09 ***
## factor(myprecinct)52 0.491898 0.055458 8.870 < 2e-16 ***
## factor(myprecinct)53 0.599687 0.058486 10.254 < 2e-16 ***
## factor(myprecinct)54 0.491840 0.060042 8.192 2.58e-16 ***
## factor(myprecinct)55 0.497511 0.059023 8.429 < 2e-16 ***
## factor(myprecinct)56 0.890877 0.065982 13.502 < 2e-16 ***
## factor(myprecinct)57 1.532782 0.061366 24.978 < 2e-16 ***
## factor(myprecinct)58 1.578846 0.054528 28.955 < 2e-16 ***
## factor(myprecinct)59 1.083726 0.058920 18.393 < 2e-16 ***
## factor(myprecinct)60 0.833828 0.054216 15.380 < 2e-16 ***
## factor(myprecinct)61 1.277336 0.056757 22.506 < 2e-16 ***
## factor(myprecinct)62 1.206378 0.056666 21.289 < 2e-16 ***
## factor(myprecinct)63 1.308782 0.058301 22.449 < 2e-16 ***
## factor(myprecinct)64 1.712497 0.057772 29.642 < 2e-16 ***
## factor(myprecinct)65 1.899204 0.055154 34.434 < 2e-16 ***
## factor(myprecinct)66 2.102128 0.054231 38.763 < 2e-16 ***
## factor(myprecinct)67 1.202841 0.054957 21.887 < 2e-16 ***
## factor(myprecinct)68 2.201257 0.059245 37.155 < 2e-16 ***
## factor(myprecinct)69 1.720700 0.058227 29.551 < 2e-16 ***

```

```
## factor(myprecinct)70 1.032004 0.054813 18.828 < 2e-16 ***
## factor(myprecinct)71 1.479415 0.054239 27.276 < 2e-16 ***
## factor(myprecinct)72 1.464985 0.053737 27.262 < 2e-16 ***
## factor(myprecinct)73 0.991018 0.053585 18.494 < 2e-16 ***
## factor(myprecinct)74 1.151460 0.058023 19.845 < 2e-16 ***
## factor(myprecinct)75 1.571225 0.075731 20.747 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 46120.3 on 224 degrees of freedom
## Residual deviance: 3427.1 on 148 degrees of freedom
## AIC: 5287.8
##
## Number of Fisher Scoring iterations: 4
```

```
(p = length(coef(model3)))
```

```
## [1] 77
```

```
(yhat = predict(model3,type="response"))
```

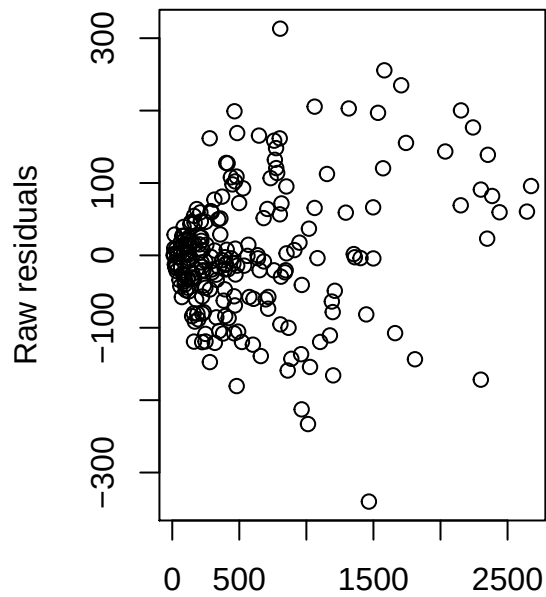
```
##      1      2      3      4      5      6
## 246.82622 75.06056 63.11321 163.39124 122.09941 61.50935
##      7      8      9     10     11     12
## 964.71248 279.28185 359.00567 398.06930 765.01575 247.91495
##     13     14     15     16     17     18
## 574.49724 646.55494 378.94782 638.99573 456.19161 201.81266
##     19     20     21     22     23     24
## 327.67874 180.39148 140.92978 965.82475 439.04586 167.12940
##     25     26     27     28     29     30
## 130.91718 87.62078 66.46204 560.69388 371.83184 223.47428
##     31     32     33     34     35     36
## 358.11247 247.22419 232.66334 467.10622 254.40297 162.49081
##     37     38     39     40     41     42
## 1019.11983 1082.82113 113.05904 601.35981 479.14907 84.49112
##     43     44     45     46     47     48
## 1446.55246 802.28952 124.15802 716.40325 209.60755 38.98920
##     49     50     51     52     53     54
## 845.58049 83.07459 17.34492 1466.09974 804.84528 88.05498
##     55     56     57     58     59     60
## 1211.91434 84.72591 14.35975 484.18141 801.38732 280.43127
##     61     62     63     64     65     66
## 328.60257 1061.52837 222.86906 1581.37857 2300.67450 147.94693
##     67     68     69     70     71     72
## 733.30315 1353.21080 247.48604 1534.09162 1027.17142 60.73696
##     73     74     75     76     77     78
## 1707.03062 1809.49947 168.46991 814.21837 696.76882 30.01280
##     79     80     81     82     83     84
## 806.32290 714.66243 484.01467 406.23207 469.71284 26.05508
##     85     86     87     88     89     90
## 2301.11472 392.53920 194.34608 850.80802 1189.74126 116.45072
##     91     92     93     94     95     96
## 651.46230 810.41768 343.12002 391.89259 498.69856 150.40885
```

	97	98	99	100	101	102
##	1175.56534	1571.69512	218.73954	1316.04943	887.90013	606.05045
##	103	104	105	106	107	108
##	203.29738	103.60919	417.09343	148.35554	331.77393	708.87053
##	109	110	111	112	113	114
##	681.51334	93.74325	221.74342	383.54808	960.52388	463.92804
##	115	116	117	118	119	120
##	2347.88391	78.11231	38.00378	93.23685	294.34877	471.41438
##	121	122	123	124	125	126
##	1661.39752	134.54882	183.05366	2035.51550	420.60185	222.88265
##	127	128	129	130	131	132
##	1497.65137	114.60038	32.74825	139.47596	1293.98046	235.54358
##	133	134	135	136	137	138
##	2644.54143	280.10835	28.35021	2152.73654	1102.58831	187.67514
##	139	140	141	142	143	144
##	536.20536	372.66178	92.13285	1499.23274	63.76391	11.00335
##	145	146	147	148	149	150
##	402.26567	519.65224	119.08209	2675.21574	350.46315	28.32110
##	151	152	153	154	155	156
##	1061.50026	123.72114	22.77860	759.54750	1196.34504	160.10747
##	157	158	159	160	161	162
##	910.68120	216.90836	88.41044	852.17531	129.82180	13.00289
##	163	164	165	166	167	168
##	287.88324	758.67183	93.44494	109.88066	294.56858	167.55076
##	169	170	171	172	173	174
##	528.37514	149.64854	177.97633	2150.84235	394.84240	125.31524
##	175	176	177	178	179	180
##	461.54424	448.97335	236.48242	2440.45694	399.48565	101.05741
##	181	182	183	184	185	186
##	289.10475	867.06856	468.82670	1403.73305	113.92469	117.34225
##	187	188	189	190	191	192
##	683.38577	347.54649	217.06775	840.40694	297.57756	213.01550
##	193	194	195	196	197	198
##	491.31697	1361.55758	437.12545	1012.02041	1153.56515	775.41445
##	199	200	201	202	203	204
##	480.51516	1742.65471	206.83014	316.35441	225.32520	567.32039
##	205	206	207	208	209	210
##	461.66803	392.45478	412.87720	2242.29080	162.10977	71.59943
##	211	212	213	214	215	216
##	1199.76603	948.41649	773.81748	861.04754	2384.97084	313.98162
##	217	218	219	220	221	222
##	2353.08599	660.09977	636.81424	317.84167	226.94030	782.21803
##	223	224	225			
##	19.39365	53.87865	248.72770			

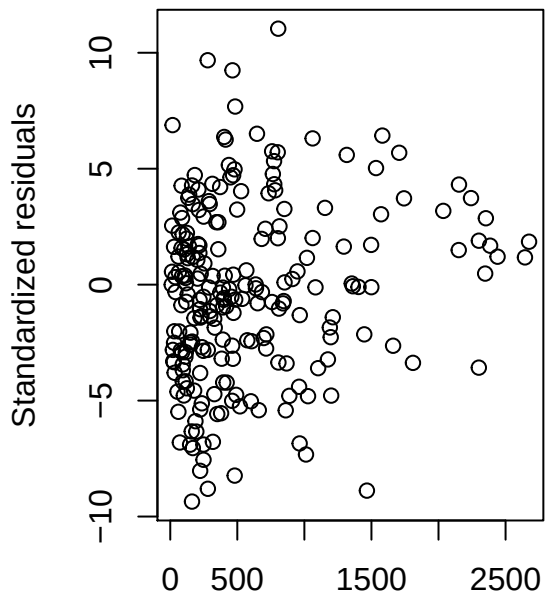
```

#residual plot
z = (mystops-yhat)/sqrt(yhat)
par(mfrow=c(1,2))
plot(yhat,mystops-yhat,xlab="Predicted values",ylab="Raw residuals")
plot(yhat,z,xlab="Predicted values",ylab="Standardized residuals")

```



Predicted values



Predicted values

```
#overdispersion ratio
sum(z^2/(n-p))
```

```
## [1] 21.88505
```

```
#p-value
1-pchisq(sum(z^2),n-p)
```

```
## [1] 0
```

```
#Negative binomial model
```

```
library(MASS)
```

```
model4 <- glm.nb(mystops ~ factor(myeth)+factor(myprecinct))
```

```
summary(model4)
```

```
##
```

```
## Call:
```

```
## glm.nb(formula = mystops ~ factor(myeth) + factor(myprecinct),
##       init.theta = 1.660407889, link = log)
```

```
##
```

```
## Deviance Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -3.2396  -0.9592  -0.2667   0.5493   1.6755
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      5.57132    0.45754  12.177 < 2e-16 ***
## factor(myeth)2     -0.63764    0.12692  -5.024 5.06e-07 ***
## factor(myeth)3     -1.58734    0.12712 -12.486 < 2e-16 ***
## factor(myprecinct)2 -0.05104    0.63892  -0.080 0.936328
## factor(myprecinct)3  1.51533    0.63679   2.380 0.017329 *
## factor(myprecinct)4  1.23613    0.63697   1.941 0.052302 .
## factor(myprecinct)5  1.45482    0.63682   2.284 0.022343 *
## factor(myprecinct)6  1.16865    0.63702   1.835 0.066572 .
```

```

## factor(myprecinct)7 0.65364 0.63756 1.025 0.305257
## factor(myprecinct)8 1.26181 0.63695 1.981 0.047589 *
## factor(myprecinct)9 -0.34399 0.63982 -0.538 0.590832
## factor(myprecinct)10 0.98744 0.63718 1.550 0.121213
## factor(myprecinct)11 0.86494 0.63731 1.357 0.174721
## factor(myprecinct)12 0.91312 0.63725 1.433 0.151890
## factor(myprecinct)13 1.48827 0.63680 2.337 0.019435 *
## factor(myprecinct)14 0.96813 0.63720 1.519 0.128676
## factor(myprecinct)15 1.55186 0.63677 2.437 0.014806 *
## factor(myprecinct)16 0.53979 0.63772 0.846 0.397308
## factor(myprecinct)17 0.41645 0.63792 0.653 0.513865
## factor(myprecinct)18 1.57448 0.63676 2.473 0.013411 *
## factor(myprecinct)19 0.65697 0.63756 1.030 0.302798
## factor(myprecinct)20 1.31738 0.63691 2.068 0.038604 *
## factor(myprecinct)21 1.38375 0.63687 2.173 0.029799 *
## factor(myprecinct)22 2.05949 0.63655 3.235 0.001215 **
## factor(myprecinct)23 1.64524 0.63672 2.584 0.009768 **
## factor(myprecinct)24 1.48372 0.63681 2.330 0.019809 *
## factor(myprecinct)25 1.93796 0.63659 3.044 0.002332 **
## factor(myprecinct)26 1.02029 0.63715 1.601 0.109303
## factor(myprecinct)27 1.86946 0.63662 2.937 0.003319 **
## factor(myprecinct)28 0.55159 0.63770 0.865 0.387055
## factor(myprecinct)29 1.53325 0.63678 2.408 0.016048 *
## factor(myprecinct)30 1.48916 0.63680 2.338 0.019361 *
## factor(myprecinct)31 1.63104 0.63673 2.562 0.010419 *
## factor(myprecinct)32 0.94903 0.63722 1.489 0.136401
## factor(myprecinct)33 1.90363 0.63661 2.990 0.002787 **
## factor(myprecinct)34 1.95996 0.63658 3.079 0.002078 **
## factor(myprecinct)35 1.10086 0.63708 1.728 0.083991 .
## factor(myprecinct)36 1.73147 0.63668 2.720 0.006537 **
## factor(myprecinct)37 0.85159 0.63732 1.336 0.181482
## factor(myprecinct)38 1.87037 0.63662 2.938 0.003304 **
## factor(myprecinct)39 1.22110 0.63698 1.917 0.055236 .
## factor(myprecinct)40 1.30204 0.63692 2.044 0.040927 *
## factor(myprecinct)41 1.36880 0.63688 2.149 0.031615 *
## factor(myprecinct)42 1.52584 0.63678 2.396 0.016567 *
## factor(myprecinct)43 0.80152 0.63738 1.258 0.208559
## factor(myprecinct)44 1.48691 0.63680 2.335 0.019546 *
## factor(myprecinct)45 1.40908 0.63685 2.213 0.026927 *
## factor(myprecinct)46 1.79390 0.63665 2.818 0.004837 **
## factor(myprecinct)47 0.80726 0.63737 1.267 0.205319
## factor(myprecinct)48 0.75920 0.63743 1.191 0.233637
## factor(myprecinct)49 0.84215 0.63733 1.321 0.186377
## factor(myprecinct)50 1.46853 0.63682 2.306 0.021107 *
## factor(myprecinct)51 0.50639 0.63777 0.794 0.427198
## factor(myprecinct)52 1.46965 0.63681 2.308 0.021009 *
## factor(myprecinct)53 0.74943 0.63744 1.176 0.239716
## factor(myprecinct)54 0.39532 0.63795 0.620 0.535473
## factor(myprecinct)55 0.93438 0.63723 1.466 0.142566
## factor(myprecinct)56 0.65944 0.63755 1.034 0.300979
## factor(myprecinct)57 0.58559 0.63765 0.918 0.358438
## factor(myprecinct)58 1.48958 0.63680 2.339 0.019327 *
## factor(myprecinct)59 1.10992 0.63707 1.742 0.081468 .
## factor(myprecinct)60 1.49026 0.63680 2.340 0.019273 *

```

```

## factor(myprecinct)61 1.74752 0.63667 2.745 0.006055 **
## factor(myprecinct)62 1.02667 0.63714 1.611 0.107100
## factor(myprecinct)63 1.14706 0.63704 1.801 0.071766 .
## factor(myprecinct)64 1.19511 0.63700 1.876 0.060635 .
## factor(myprecinct)65 1.96405 0.63658 3.085 0.002033 **
## factor(myprecinct)66 2.25959 0.63649 3.550 0.000385 ***
## factor(myprecinct)67 1.85246 0.63663 2.910 0.003616 **
## factor(myprecinct)68 1.50402 0.63679 2.362 0.018183 *
## factor(myprecinct)69 1.54453 0.63677 2.426 0.015285 *
## factor(myprecinct)70 1.18408 0.63701 1.859 0.063054 .
## factor(myprecinct)71 2.23469 0.63650 3.511 0.000447 ***
## factor(myprecinct)72 2.22321 0.63650 3.493 0.000478 ***
## factor(myprecinct)73 2.12357 0.63653 3.336 0.000849 ***
## factor(myprecinct)74 1.84975 0.63663 2.906 0.003666 **
## factor(myprecinct)75 0.62762 0.63760 0.984 0.324937
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.6604) family taken to be 1)
##
## Null deviance: 462.35  on 224  degrees of freedom
## Residual deviance: 246.90  on 148  degrees of freedom
## AIC: 3313.4
##
## Number of Fisher Scoring iterations: 1
##
##
## Theta: 1.660
## Std. Err.: 0.145
##
## 2 x log-likelihood: -3157.360
#####
data <- read.table("policestops.txt",header = TRUE)
attach(data)

## The following objects are masked _by_ .GlobalEnv:
##
## myarrests, myeth, myprecinct, mystops
##
## The following objects are masked from data (pos = 4):
##
## myarrests, myeth, myprecinct, mystops
model1 <- glm(mystops ~ factor(myeth)+factor(myprecinct),family=poisson(link = "log"),offset=log(myarrests))
summary(model1)

##
## Call:
## glm(formula = mystops ~ factor(myeth) + factor(myprecinct), family = poisson(link = "log"),
## offset = log(myarrests))
##
## Deviance Residuals:
## Min 1Q Median 3Q Max
## -11.1396 -3.0895 -0.1934 2.0988 10.4185
##

```

```

## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.378868   0.051019 -27.027 < 2e-16 ***
## factor(myeth)2    0.010188   0.006802   1.498 0.134190
## factor(myeth)3   -0.419001   0.009435 -44.409 < 2e-16 ***
## factor(myprecinct)2 -0.149050   0.074030 -2.013 0.044077 *
## factor(myprecinct)3  0.559955   0.056758   9.866 < 2e-16 ***
## factor(myprecinct)4  1.210636   0.057549  21.037 < 2e-16 ***
## factor(myprecinct)5  0.282865   0.056794   4.981 6.34e-07 ***
## factor(myprecinct)6  1.144204   0.058047  19.712 < 2e-16 ***
## factor(myprecinct)7  0.218173   0.064335   3.391 0.000696 ***
## factor(myprecinct)8 -0.390565   0.056868 -6.868 6.51e-12 ***
## factor(myprecinct)9  0.485112   0.078151   6.207 5.39e-10 ***
## factor(myprecinct)10 0.417068   0.058851   7.087 1.37e-12 ***
## factor(myprecinct)11 0.654417   0.061586  10.626 < 2e-16 ***
## factor(myprecinct)12 1.148698   0.061067  18.811 < 2e-16 ***
## factor(myprecinct)13 1.053509   0.055250  19.068 < 2e-16 ***
## factor(myprecinct)14 0.614115   0.058803  10.444 < 2e-16 ***
## factor(myprecinct)15 1.054904   0.054956  19.195 < 2e-16 ***
## factor(myprecinct)16 0.797715   0.060290  13.231 < 2e-16 ***
## factor(myprecinct)17 -0.049312   0.060478 -0.815 0.414864
## factor(myprecinct)18 0.131645   0.054981   2.394 0.016649 *
## factor(myprecinct)19 0.319164   0.057999   5.503 3.74e-08 ***
## factor(myprecinct)20 -0.040574   0.056928 -0.713 0.476016
## factor(myprecinct)21 0.410988   0.056805   7.235 4.65e-13 ***
## factor(myprecinct)22 1.211227   0.053399  22.683 < 2e-16 ***
## factor(myprecinct)23 0.599919   0.055064  10.895 < 2e-16 ***
## factor(myprecinct)24 1.319454   0.054600  24.166 < 2e-16 ***
## factor(myprecinct)25 0.913359   0.053596  17.041 < 2e-16 ***
## factor(myprecinct)26 -0.147614   0.057006 -2.589 0.009612 **
## factor(myprecinct)27 1.895485   0.055664  34.052 < 2e-16 ***
## factor(myprecinct)28 -0.764964   0.060915 -12.558 < 2e-16 ***
## factor(myprecinct)29 1.124710   0.054267  20.726 < 2e-16 ***
## factor(myprecinct)30 0.524652   0.055374   9.475 < 2e-16 ***
## factor(myprecinct)31 1.649498   0.056167  29.368 < 2e-16 ***
## factor(myprecinct)32 1.386276   0.059679  23.229 < 2e-16 ***
## factor(myprecinct)33 1.083192   0.054214  19.980 < 2e-16 ***
## factor(myprecinct)34 1.518972   0.054355  27.945 < 2e-16 ***
## factor(myprecinct)35 0.879407   0.063190  13.917 < 2e-16 ***
## factor(myprecinct)36 1.616306   0.058796  27.490 < 2e-16 ***
## factor(myprecinct)37 1.409430   0.060008  23.487 < 2e-16 ***
## factor(myprecinct)38 1.763813   0.056195  31.388 < 2e-16 ***
## factor(myprecinct)39 0.501081   0.054841   9.137 < 2e-16 ***
## factor(myprecinct)40 1.531984   0.061467  24.924 < 2e-16 ***
## factor(myprecinct)41 1.913898   0.055716  34.351 < 2e-16 ***
## factor(myprecinct)42 1.084217   0.054511  19.890 < 2e-16 ***
## factor(myprecinct)43 0.541787   0.056645   9.565 < 2e-16 ***
## factor(myprecinct)44 0.883038   0.056671  15.582 < 2e-16 ***
## factor(myprecinct)45 0.838439   0.054215  15.465 < 2e-16 ***
## factor(myprecinct)46 0.655855   0.053750  12.202 < 2e-16 ***
## factor(myprecinct)47 1.128144   0.059981  18.808 < 2e-16 ***
## factor(myprecinct)48 0.411880   0.056895   7.239 4.51e-13 ***
## factor(myprecinct)49 1.101219   0.059681  18.452 < 2e-16 ***
## factor(myprecinct)50 0.878797   0.054109  16.241 < 2e-16 ***

```

```

## factor(myprecinct)51 0.348980 0.058550 5.960 2.52e-09 ***
## factor(myprecinct)52 0.491898 0.055458 8.870 < 2e-16 ***
## factor(myprecinct)53 0.599687 0.058486 10.254 < 2e-16 ***
## factor(myprecinct)54 0.491840 0.060042 8.192 2.58e-16 ***
## factor(myprecinct)55 0.497511 0.059023 8.429 < 2e-16 ***
## factor(myprecinct)56 0.890877 0.065982 13.502 < 2e-16 ***
## factor(myprecinct)57 1.532782 0.061366 24.978 < 2e-16 ***
## factor(myprecinct)58 1.578846 0.054528 28.955 < 2e-16 ***
## factor(myprecinct)59 1.083726 0.058920 18.393 < 2e-16 ***
## factor(myprecinct)60 0.833828 0.054216 15.380 < 2e-16 ***
## factor(myprecinct)61 1.277336 0.056757 22.506 < 2e-16 ***
## factor(myprecinct)62 1.206378 0.056666 21.289 < 2e-16 ***
## factor(myprecinct)63 1.308782 0.058301 22.449 < 2e-16 ***
## factor(myprecinct)64 1.712497 0.057772 29.642 < 2e-16 ***
## factor(myprecinct)65 1.899204 0.055154 34.434 < 2e-16 ***
## factor(myprecinct)66 2.102128 0.054231 38.763 < 2e-16 ***
## factor(myprecinct)67 1.202841 0.054957 21.887 < 2e-16 ***
## factor(myprecinct)68 2.201257 0.059245 37.155 < 2e-16 ***
## factor(myprecinct)69 1.720700 0.058227 29.551 < 2e-16 ***
## factor(myprecinct)70 1.032004 0.054813 18.828 < 2e-16 ***
## factor(myprecinct)71 1.479415 0.054239 27.276 < 2e-16 ***
## factor(myprecinct)72 1.464985 0.053737 27.262 < 2e-16 ***
## factor(myprecinct)73 0.991018 0.053585 18.494 < 2e-16 ***
## factor(myprecinct)74 1.151460 0.058023 19.845 < 2e-16 ***
## factor(myprecinct)75 1.571225 0.075731 20.747 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 46120.3 on 224 degrees of freedom
## Residual deviance: 3427.1 on 148 degrees of freedom
## AIC: 5287.8
##
## Number of Fisher Scoring iterations: 4
model2 <- glm(mystops ~ factor(myprecinct),family=poisson(link = "log"),offset=log(myarrests))
summary(model2)

##
## Call:
## glm(formula = mystops ~ factor(myprecinct), family = poisson(link = "log"),
##      offset = log(myarrests))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1766  -3.7591  -0.6493   2.1211  11.1977
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.45892    0.05096  -28.626 < 2e-16 ***
## factor(myprecinct)2  -0.15397    0.07402   -2.080 0.037516 *
## factor(myprecinct)3   0.53134    0.05676   9.362 < 2e-16 ***
## factor(myprecinct)4   1.20823    0.05750  21.013 < 2e-16 ***
## factor(myprecinct)5   0.25033    0.05677   4.410 1.03e-05 ***

```

```

## factor(myprecinct)6 1.14969 0.05804 19.809 < 2e-16 ***
## factor(myprecinct)7 0.19369 0.06433 3.011 0.002605 **
## factor(myprecinct)8 -0.36168 0.05686 -6.360 2.01e-10 ***
## factor(myprecinct)9 0.45340 0.07814 5.802 6.55e-09 ***
## factor(myprecinct)10 0.40422 0.05884 6.869 6.44e-12 ***
## factor(myprecinct)11 0.60212 0.06157 9.780 < 2e-16 ***
## factor(myprecinct)12 1.14005 0.06106 18.670 < 2e-16 ***
## factor(myprecinct)13 1.11218 0.05522 20.142 < 2e-16 ***
## factor(myprecinct)14 0.66114 0.05879 11.247 < 2e-16 ***
## factor(myprecinct)15 1.11143 0.05494 20.229 < 2e-16 ***
## factor(myprecinct)16 0.85911 0.06028 14.252 < 2e-16 ***
## factor(myprecinct)17 0.02212 0.06045 0.366 0.714403
## factor(myprecinct)18 0.19585 0.05497 3.563 0.000366 ***
## factor(myprecinct)19 0.39418 0.05797 6.800 1.05e-11 ***
## factor(myprecinct)20 -0.04488 0.05689 -0.789 0.430175
## factor(myprecinct)21 0.42784 0.05672 7.543 4.60e-14 ***
## factor(myprecinct)22 1.27804 0.05334 23.959 < 2e-16 ***
## factor(myprecinct)23 0.63184 0.05501 11.486 < 2e-16 ***
## factor(myprecinct)24 1.39145 0.05458 25.495 < 2e-16 ***
## factor(myprecinct)25 0.97477 0.05356 18.199 < 2e-16 ***
## factor(myprecinct)26 -0.07310 0.05698 -1.283 0.199480
## factor(myprecinct)27 1.86040 0.05564 33.435 < 2e-16 ***
## factor(myprecinct)28 -0.69462 0.06088 -11.410 < 2e-16 ***
## factor(myprecinct)29 1.17167 0.05426 21.595 < 2e-16 ***
## factor(myprecinct)30 0.58244 0.05533 10.527 < 2e-16 ***
## factor(myprecinct)31 1.63936 0.05614 29.203 < 2e-16 ***
## factor(myprecinct)32 1.39835 0.05965 23.443 < 2e-16 ***
## factor(myprecinct)33 1.13076 0.05417 20.874 < 2e-16 ***
## factor(myprecinct)34 1.49552 0.05434 27.520 < 2e-16 ***
## factor(myprecinct)35 0.69834 0.06308 11.071 < 2e-16 ***
## factor(myprecinct)36 1.42827 0.05864 24.357 < 2e-16 ***
## factor(myprecinct)37 1.38081 0.06000 23.012 < 2e-16 ***
## factor(myprecinct)38 1.72327 0.05613 30.702 < 2e-16 ***
## factor(myprecinct)39 0.57345 0.05480 10.464 < 2e-16 ***
## factor(myprecinct)40 1.36350 0.06133 22.232 < 2e-16 ***
## factor(myprecinct)41 1.94759 0.05570 34.964 < 2e-16 ***
## factor(myprecinct)42 1.12341 0.05450 20.611 < 2e-16 ***
## factor(myprecinct)43 0.61223 0.05662 10.814 < 2e-16 ***
## factor(myprecinct)44 0.89956 0.05654 15.911 < 2e-16 ***
## factor(myprecinct)45 0.91446 0.05419 16.877 < 2e-16 ***
## factor(myprecinct)46 0.71109 0.05374 13.232 < 2e-16 ***
## factor(myprecinct)47 1.16501 0.05997 19.427 < 2e-16 ***
## factor(myprecinct)48 0.48871 0.05686 8.595 < 2e-16 ***
## factor(myprecinct)49 1.12823 0.05965 18.914 < 2e-16 ***
## factor(myprecinct)50 0.95519 0.05408 17.662 < 2e-16 ***
## factor(myprecinct)51 0.42029 0.05853 7.181 6.90e-13 ***
## factor(myprecinct)52 0.53885 0.05541 9.725 < 2e-16 ***
## factor(myprecinct)53 0.64434 0.05848 11.018 < 2e-16 ***
## factor(myprecinct)54 0.56642 0.06002 9.437 < 2e-16 ***
## factor(myprecinct)55 0.54228 0.05895 9.200 < 2e-16 ***
## factor(myprecinct)56 0.83358 0.06592 12.645 < 2e-16 ***
## factor(myprecinct)57 1.51169 0.06136 24.634 < 2e-16 ***
## factor(myprecinct)58 1.63623 0.05451 30.015 < 2e-16 ***
## factor(myprecinct)59 1.06544 0.05890 18.089 < 2e-16 ***

```

```

## factor(myprecinct)60 0.89751 0.05420 16.560 < 2e-16 ***
## factor(myprecinct)61 1.22220 0.05668 21.563 < 2e-16 ***
## factor(myprecinct)62 1.25044 0.05665 22.074 < 2e-16 ***
## factor(myprecinct)63 1.30477 0.05830 22.381 < 2e-16 ***
## factor(myprecinct)64 1.71575 0.05777 29.699 < 2e-16 ***
## factor(myprecinct)65 1.89004 0.05508 34.313 < 2e-16 ***
## factor(myprecinct)66 2.05709 0.05420 37.955 < 2e-16 ***
## factor(myprecinct)67 1.24653 0.05485 22.725 < 2e-16 ***
## factor(myprecinct)68 2.04688 0.05915 34.603 < 2e-16 ***
## factor(myprecinct)69 1.64678 0.05820 28.298 < 2e-16 ***
## factor(myprecinct)70 1.09777 0.05478 20.038 < 2e-16 ***
## factor(myprecinct)71 1.43324 0.05422 26.435 < 2e-16 ***
## factor(myprecinct)72 1.50667 0.05365 28.083 < 2e-16 ***
## factor(myprecinct)73 0.98584 0.05359 18.398 < 2e-16 ***
## factor(myprecinct)74 0.96527 0.05789 16.675 < 2e-16 ***
## factor(myprecinct)75 1.31457 0.07552 17.407 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 46120.3 on 224 degrees of freedom
## Residual deviance: 5896.3 on 150 degrees of freedom
## AIC: 7752.9
##
## Number of Fisher Scoring iterations: 4
model3 <- glm(mystops ~ factor(myeth),family=poisson(link = "log"),offset=log(myarrests))
summary(model3)

##
## Call:
## glm(formula = mystops ~ factor(myeth), family = poisson(link = "log"),
## offset = log(myarrests))
##
## Deviance Residuals:
## Min 1Q Median 3Q Max
## -47.327 -7.740 -0.182 10.241 39.140
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.588086 0.003784 -155.40 <2e-16 ***
## factor(myeth)2 0.070208 0.006061 11.58 <2e-16 ***
## factor(myeth)3 -0.161581 0.008558 -18.88 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 46120 on 224 degrees of freedom
## Residual deviance: 45437 on 222 degrees of freedom
## AIC: 47150
##
## Number of Fisher Scoring iterations: 5

```

```

model4 <- glm(mystops ~ 1,family=poisson(link = "log"),offset=log(myarrests))
summary(model4)

##
## Call:
## glm(formula = mystops ~ 1, family = poisson(link = "log"), offset = log(myarrests))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -46.702   -8.437    0.020   10.132   40.795
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.587715   0.002758  -213.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 46120  on 224  degrees of freedom
## Residual deviance: 46120  on 224  degrees of freedom
## AIC: 47829
##
## Number of Fisher Scoring iterations: 5
#####

ratio1 = rep(0,75)
ratio2 = rep(0.75)

for(i in 1:75)
{
  ratio1[i] = myarrests[(myprecinct==i)&(myeth==2)]*exp(coef(model1)["factor(myeth)2"])/myarrests[(myprecinct==i)&(myeth==2)]
  ratio2[i] = myarrests[(myprecinct==i)&(myeth==3)]*exp(coef(model1)["factor(myeth)3"])/myarrests[(myprecinct==i)&(myeth==3)]
}

plot.new()
par(mfrow=c(1,1))

```

```
plot(1:75, ratio2, type="l", lwd=2, xlab="Precinct", ylab="Ratio of expected stops")  
lines(1:75, ratio1, lwd=2, lty="dotted")  
abline(h=1)
```

